Worksheet 7.6

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Chemical reactions: Summary

Answers

Use the words listed below to complete the sentences.

acidic	antacids	basic	oxidation
combustion	oxygen	dioxide	equations
melting	exothermic	neutralise	nitrogen
products	reactants	reversible	endothermic

1. In a chemical change one or more new substances are formed and the process is not easily reversible.
2. The burning of a candle involves both physical and chemical changes. The melting of the wax is a physical change and the combustion of the wax vapour is a chemical change.
3. In a chemical reaction the starting materials are called the reactants and the substances that are formed are called the products.
4. Word equations can be used to describe the chemical change.
5. Exothermic reactions produce heat.
6. Endothermic reactions absorb heat from their surroundings.
7. Acids are substances that neutralise bases to produce salts and water. Indicators can be used to determine whether a substance is an acid or a base.
8. The pH scale is used to classify acids and bases. If the pH is less than 7, then the substance is acidic If the pH is above 7, then the substance is basic.
9. Acidic gases such as sulfur dioxide and nitrogen dioxide produce acid rain when they dissolve in rain. Acid rain can cause significant environmental destruction.
10. Indigestion can be relieved by using antacids, which neutralise excess acidity in the stomach.
11. When a fuel burns in the presence of air or oxygen heat is produced. If the fuel is a hydrocarbon, then carbon dioxide and water are formed unless the oxygen supply is limited.
12. Oxidation reactions involve the transfer of electrons from a reactant.
13. Not all oxidation reactions involve oxygen.

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